BIONEXUS STUDIO AI

BioNexus Studio AI — Complete Codebase Audit

Comprehensive Codebase Audit Report

AUDIT EXECUTIVE SUMMARY

This audit inspects the full BioNexus Studio AI codebase (React 19 + Vite 8). Key finding: The application is 100% client-side (no backend server). Scientific modules (MSA, Phylogenetics, Sequence Analysis, NCBI Explorer, 3D Protein Viewer) possess genuine client-side algorithms, but BLAST is mocked locally against 4 genes, user auth is simulated in localStorage with Base64 passwords, the AI Assistant is unmounted/orphaned, and LabUtilities has runtime crashing bugs.

1. PROJECT DIRECTORY STRUCTURE

- **Root Level:** Configured with Vite 8 (vite.config.js), React 19 (package.json), Firebase hosting rules (firebase.json), and Oxlint (.oxlintrc.json).
- **public/:** Static assets (robots.txt, sitemap.xml, _redirects for SPA routing, favicon.svg, and og-image.png).
- **src/context/:** WorkspaceContext.jsx - Provides centralized active sequence memory and project isolation across all components.
- **src/services/:** authService.js (localStorage user accounts) and dataService.js (storage partitioning).
- **src/utils/:** 12 core utility engines: sequenceEngine, primerEngine, blastEngine, alignmentEngine, phyloEngine, ncbiService, labCalculators, sequenceValidator, pdfGenerator, analytics, seo, aiAssistantEngine.
- **src/components/:** 20 React UI components covering 10 scientific tabs and modal management dialogs.

2. FRONTEND FRAMEWORK & PROGRAMMING LANGUAGES

- **Framework:** React 19 (v19.2.8) and React-DOM (v19.2.8) in package.json.
- **Build Tooling:** Vite (v8.3.0) with @vitejs/plugin-react (v6.1.1). Fast HMR dev server and optimized rollup bundle.
- **Languages:** JavaScript (ES Modules + JSX) and Vanilla CSS. No TypeScript (files are .js/.jsx) and no TailwindCSS.
- **Libraries:** Lucide React (v1.47.0) for icons; 3Dmol.js (v2.5.5) for WebGL PDB protein models; jsPDF (v4.2.1) & html2canvas (v1.4.1) for PDF reporting.

3. BACKEND INFRASTRUCTURE ASSESSMENT

NO BACKEND SERVER EXISTS

There is no Node.js/Express, Python/FastAPI, Go, or serverless functions folder in this repository. The entire platform runs 100% inside the user browser. While .env.example references VITE_API_BASE_URL=https://api.bionexusstudio.ai, this variable is never imported or used in the source code.

4. HOSTING & DEPLOYMENT CONFIGURATION

- **Firebase Hosting:** firebase.json specifies "public": "dist" with a SPA rewrite "**" -> "/index.html". .firebaserc binds to project "bionexux-bbc9b".
- **Live Domains:** Default domain https://bionexux-bbc9b.web.app configured in .env and .env.production. Custom domain https://bionexusstudio.ai set as canonical URL.
- **Netlify / Cloudflare:** public/_redirects contains "/* /index.html 200" for seamless SPA client-side routing on edge static platforms.

5. EXTERNAL APIS & AI SERVICES USAGE

- **NCBI E-Utilities:** Calls eutils.ncbi.nlm.nih.gov/entrez/eutils (esearch, esummary, efetch) via ncbiService.js. Direct browser fetch without API key.
- **RCSB PDB:** Downloads 3D coordinate files (.pdb) from files.rcsb.org and renders them via 3Dmol.js in ProteinStructure.jsx.
- **Google Analytics 4:** analytics.js injects gtag.js in production (Measurement ID: G-81PM8QDEPV).

- **AI Services Reality:** NO remote LLM API is used (no OpenAI, no Gemini, no Anthropic). The AI assistant uses offline regex keyword pattern matching with static biological templates.

6. DATABASE, AUTHENTICATION & STORAGE AUDIT

- **Database Status:** NO database (no SQL, MongoDB, Firestore, or Supabase). All data is saved in browser localStorage (bionexus_active_sequence, bionexus_primers, etc.).
- **Authentication Status:** SIMULATED in localStorage. authService.js encodes passwords with btoa() (Base64). Passwords are not encrypted or hashed. Reset password does not send emails.
- **Cloud Storage Status:** NO cloud storage. Workspace backup/restore is handled via client-side JSON files; PDF reports are rendered locally using jsPDF.

7. MAJOR MODULES & FILE RESPONSIBILITY MATRIX

- **Primer Designer:** src/components/PrimerDesigner.jsx & src/utils/primerEngine.js. Forward/Reverse primer generation, Tm, hairpins, dimers.
- **NCBI Explorer:** src/components/NcbiExplorer.jsx & src/utils/ncbiService.js. Entrez database search & FASTA sequence import.
- **BLAST Center:** src/components/BlastCenter.jsx & src/utils/blastEngine.js. Local Smith-Waterman alignment against 4 reference genes.
- **Sequence Analysis:** src/components/SequenceAnalysis.jsx & src/utils/sequenceEngine.js. 6-frame ORF finder, translation, MW, and isoelectric point titration.
- **Alignment Hub:** src/components/AlignmentHub.jsx & src/utils/alignmentEngine.js. Progressive Needleman-Wunsch Multiple Sequence Alignment (MSA).
- **Phylogenetics:** src/components/Phylogenetics.jsx & src/utils/phyloEngine.js. Distance models (JC69/K80/p-dist), NJ & UPGMA tree generation, SVG render, Newick export.
- **Protein Structure:** src/components/ProteinStructure.jsx. 3D WebGL rendering of PDB structures via 3Dmol.js.
- **Lab Utilities:** src/components/LabUtilities.jsx & src/utils/labCalculators.js. Master Mix, Dilutions, Molarity, Media recipes.
- **Global Workspace:** src/context/WorkspaceContext.jsx & src/components/ActiveSequenceBar.jsx. Synchronizes active sequence across all tools.
- **AI Assistant:** src/components/AiAssistant.jsx & src/utils/aiAssistantEngine.js. Currently unmounted / orphaned.

8. SCIENTIFIC ALGORITHMS IMPLEMENTED

- **Smith-Waterman:** Dynamic programming for local pairwise alignment. Computes Karlin-Altschul parameters for Bit Score and E-values (blastEngine.js).
- **Needleman-Wunsch:** Dynamic programming for end-to-end global alignment with match/mismatch scores and gap penalties (alignmentEngine.js).
- **Progressive MSA:** Sequential star alignment using the first sequence as guide, followed by consensus and conservation scoring (alignmentEngine.js).
- **Distance Models:** p-Distance (Hamming), Jukes-Cantor 1969 (JC69 logarithmic saturation correction), Kimura 2-Parameter (K80 transition/transversion ratio) (phyloEngine.js).
- **Phylogenetic Trees:** Neighbor-Joining (Saitou & Nei 1987) with Q-matrix and branch lengths; UPGMA agglomerative clustering; Newick export (phyloEngine.js).
- **Primer Thermodynamics:** Wallace rule (<14 bp), salt-adjusted empirical Tm, hairpin stem-loop search, dimer complementarity scan (primerEngine.js).
- **Protein Chemistry:** Isoelectric point (pI) solved via Henderson-Hasselbalch charge balance from pH 0 to 14; Kyte-Doolittle hydropathy profile (sequenceEngine.js).

9. FEATURE IMPLEMENTATION STATUS

- **Fully Implemented:** Sequence Analysis Suite, Phylogenetics Suite, Multiple Sequence Alignment, NCBI Explorer, 3D Protein

Structure Viewer, DNA/FASTA Validator, Theme Switching.

- **Partially Implemented:** Primer Designer (empirical Tm vs SantaLucia nearest-neighbor); Project Manager & Sequence Library (local only, no cloud sync); Single Page App Routing (history pushState).
- **Mocked / Simulated:** BLAST Center (matches 4 genes in memory instead of NCBI GenBank); Authentication (localStorage Base64 accounts, no server check).
- **Orphaned / Broken:** AiAssistant.jsx (not rendered in UI); LabUtilities.jsx (crashing bugs in PCR and Molarity tabs); SequenceConflictModal.jsx (never imported).

10. SECURITY, ACCURACY & RELIABILITY DEFECTS

CRITICAL BUG 1: LabUtilities PCR Tab Crash

LabUtilities.jsx line 98 calls pcrResult.components.map(), but labCalculators.js returns { recipe: [...] }. Because .components is undefined, the PCR tab throws an immediate uncaught TypeError on mount.

CRITICAL BUG 2: LabUtilities Molarity Tab Crash

LabUtilities.jsx lines 148-156 bind to molarity, volMl, and mw. None of these variables exist in useState (which declared molMass, desiredConc, desiredVol), causing an immediate ReferenceError.

SECURITY WARNING: Insecure Password Storage

authService.js stores passwords as passwordHash: btoa(password). Base64 encoding is not encryption or hashing; it is completely reversible in 1 second using atob().

SCIENTIFIC ACCURACY: 4-Gene BLAST Limit

BLAST Center searches only 4 hardcoded genes (BRCA1, TP53, SARS-CoV-2, ACTB). Querying other genes produces 0 hits, which can mislead users into believing their sequence does not exist in nature.

11. LEARNING PLAN & RECOMMENDED ROADMAP

- **Phase 1 (Bug Fixes):** Fix LabUtilities.jsx property mappings and state variables; mount or cleanly remove AiAssistant.jsx; clean up orphaned components.
- **Phase 2 (Bioinformatics):** Study dynamic programming in alignmentEngine.js (Needleman-Wunsch); study distance matrix Q-calculation in phyloEngine.js (Neighbor-Joining).
- **Phase 3 (Production Cloud):** Replace localStorage auth with Firebase Authentication or Supabase; replace local BLAST with NCBI QBLAST API; connect AiAssistant to an authentic generative AI API.